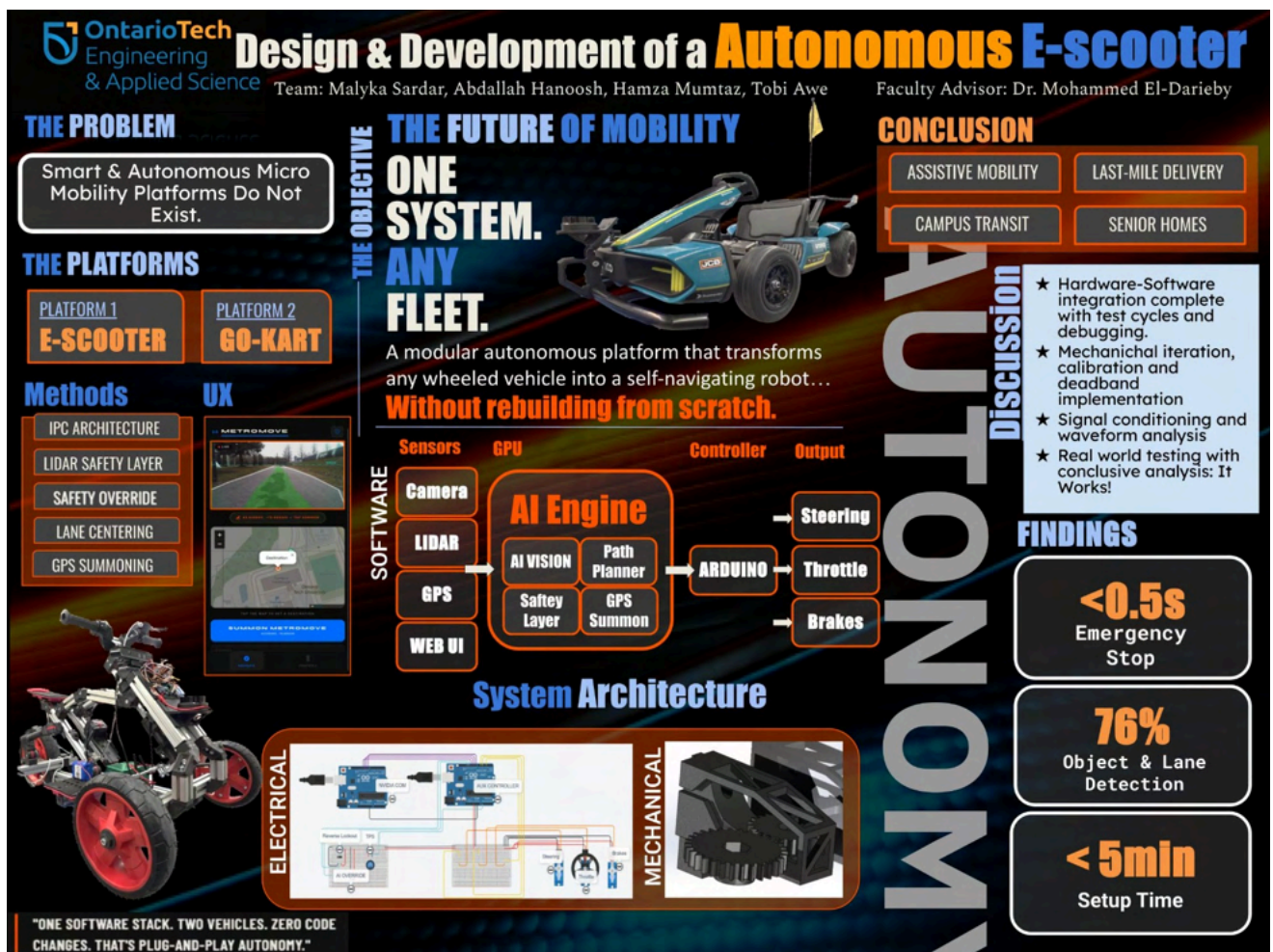


Capstone Project — Autonomous E-Scooter

Embedded & Autonomous Systems Integration



Project Snapshot

- Team: 4 (2 Electrical, 2 Software)
- Focus: Embedded control systems, CAN communication, system integration
- Scope: Autonomous electric scooter platform

My Ownership

I was responsible for the embedded and hardware-side system integration and served as the primary interface between electrical hardware and software development.

- Embedded control architecture design and implementation
- CAN-based communication development, addressing, and validation
- Sensor and actuator integration (selection, wiring, interfacing, timing)
- Hardware–software integration across subsystems
- System-level testing and debugging under real operating conditions

Key Technical Challenges

- Achieving reliable CAN communication across multiple subsystems while managing message timing and fault diagnosis
- Integrating heterogeneous sensors and actuators while maintaining signal integrity, latency constraints, and control-loop stability
- Debugging failures that spanned hardware, firmware, and software boundaries

Engineering Summary

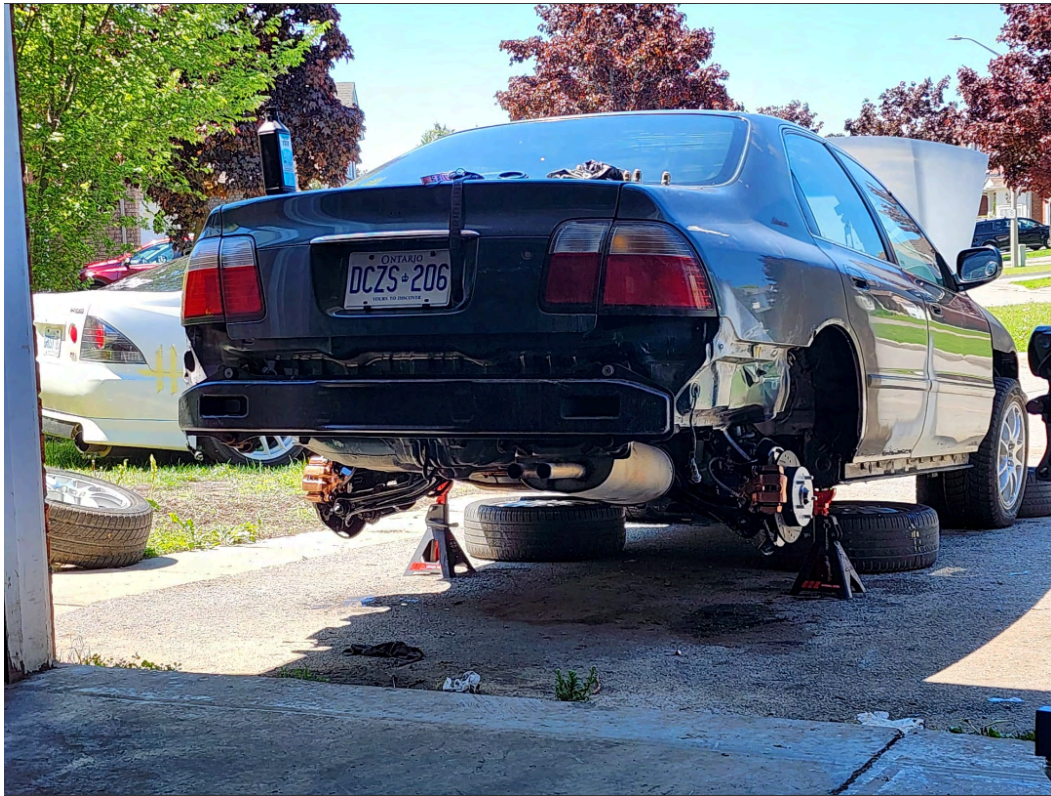
This project required approaching the system as a whole rather than as isolated components. Many integration issues only became visible once hardware, firmware, and software were operating together, requiring iterative debugging and close coordination with the software team. My role focused on identifying system-level failure points, validating communication and control behavior, and ensuring that embedded hardware constraints were clearly understood and reflected in the overall system design.

Tools & Technologies

CAN bus, embedded microcontrollers, sensors & actuators, hardware integration, system testing & validation

Honda Accord — (JTCC COMPLIANT)

Vehicle Systems Engineering & Embedded Controls



Project Snapshot

- **Platform:** 1996 Honda Accord (CD5)
- **Focus:** Embedded engine management, sensor integration, vehicle systems engineering
- **Scope:** Long-term, self-directed vehicle development project inspired by JTCC touring car platforms

My Ownership

I am responsible for the complete system design, integration, and validation of the vehicle's mechanical, electrical, and embedded subsystems. This project is self-directed and treated as a professional-grade engineering effort rather than a cosmetic or hobby build.

- Embedded engine management system design and implementation
- Custom sensor integration for engine position, pressure, and vehicle telemetry
- ECU configuration, wiring architecture, and signal validation
- Mechanical–electrical co-design of engine and drivetrain modifications
- System-level testing, troubleshooting, and iterative refinement

Key Technical Challenges

- Designing reliable **engine position sensing** using custom cam/crank solutions and retractor geometry
- Integrating aftermarket and custom sensors while maintaining signal integrity and timing accuracy
- Coordinating mechanical modifications (engine internals, balance shaft delete) with embedded control requirements
- Debugging system-level faults where failures could originate from sensors, wiring, ECU configuration, or mechanical behavior

Engineering Summary

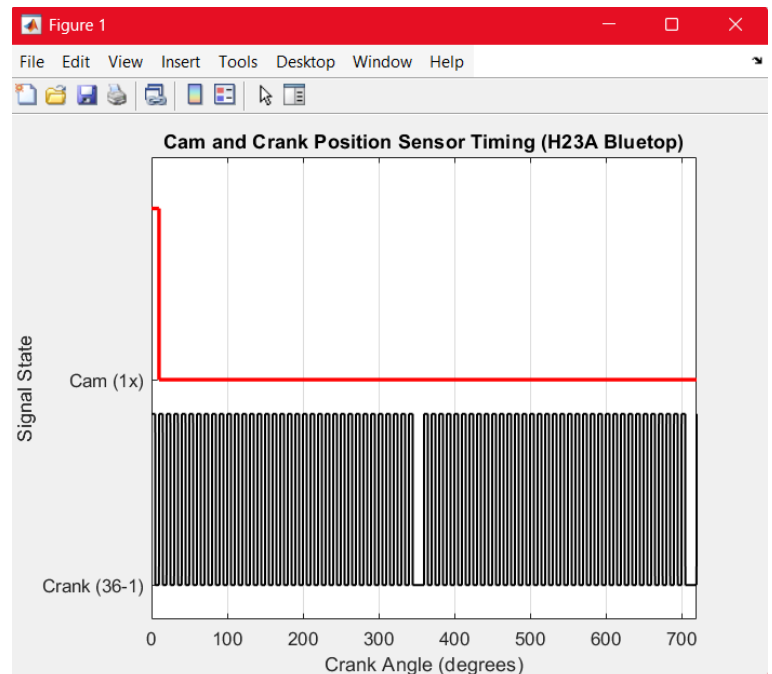
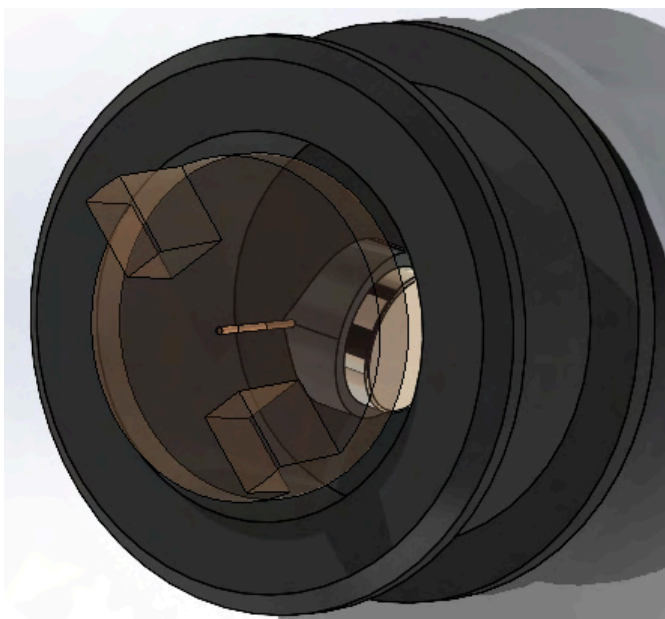
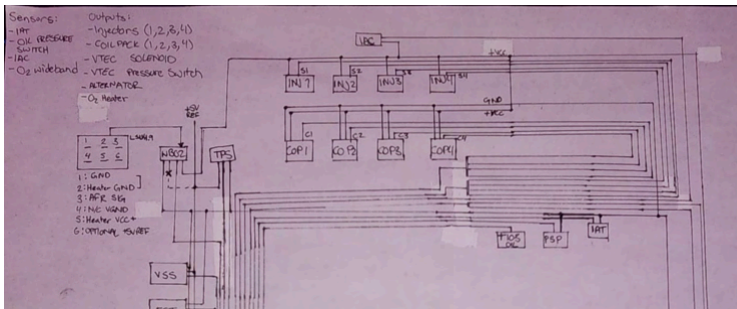
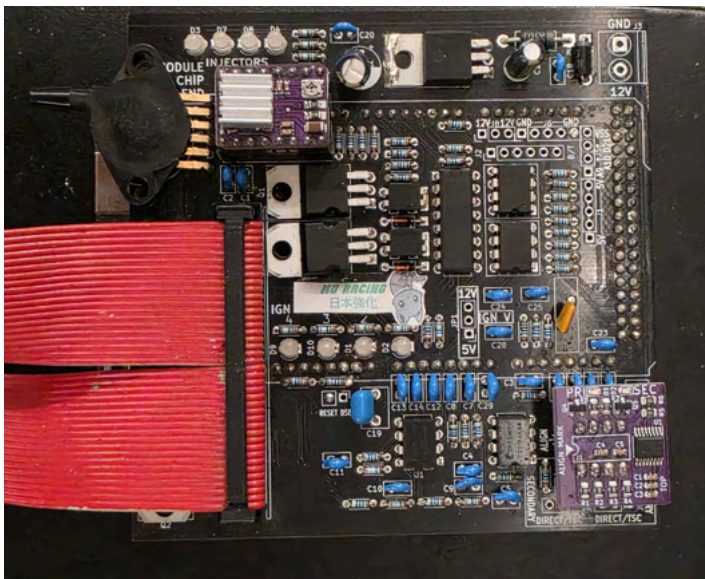
This project is centered on treating a production vehicle as a complete engineered system rather than a collection of independent modifications. Many challenges only emerge when mechanical changes, sensors, and embedded control logic interact in real operating conditions. My work focuses on designing sensing and control solutions that are electrically robust, mechanically compliant, and safe for real-world operation, while validating system behavior through structured testing and iteration.

Tools & Technologies

Embedded ECU, custom wiring harnesses, engine position sensors, CAN-based telemetry, mechanical design and fabrication, system testing & validation

Custom ECU & Engine Management Systems

Embedded Systems, Sensors & Timing-Critical Control



Project Snapshot

Focus: Embedded engine management, timing-critical sensing, hardware integration

Scope: Custom engine management and sensing system for performance automotive applications

My Ownership

I was responsible for the complete design, implementation, and validation of a custom engine management and sensing system, with an emphasis on reliability, timing accuracy, and real-world operating constraints.

- Camshaft and crankshaft position sensing design and implementation
- Custom reluctor wheel design to achieve required timing resolution and sensor compatibility
- Integration of engine instrumentation, including boost pressure measurement and RPM monitoring
- ECU configuration, wiring architecture, and signal conditioning
- Sensor calibration and validation under real operating conditions
- Development of telemetry and data logging concepts to support diagnostics and system analysis

Key Technical Challenges

Designing a reliable engine management system required solving timing-critical challenges where small errors in sensing or signal quality could directly impact system behavior.

- Achieving accurate and repeatable cam/crank position sensing through custom reluctor geometry and sensor placement
- Managing electrical noise and interference while preserving signal integrity in an automotive environment
- Synchronizing multiple sensor inputs to support stable RPM and timing calculations
- Debugging failures that could originate from sensors, wiring, ECU configuration, or calibration assumptions

Engineering Summary

This project required treating engine management as a tightly coupled embedded system rather than a collection of independent sensors. Many issues only became visible once mechanical design, sensor behavior, wiring, and embedded configuration interacted under real operating conditions.

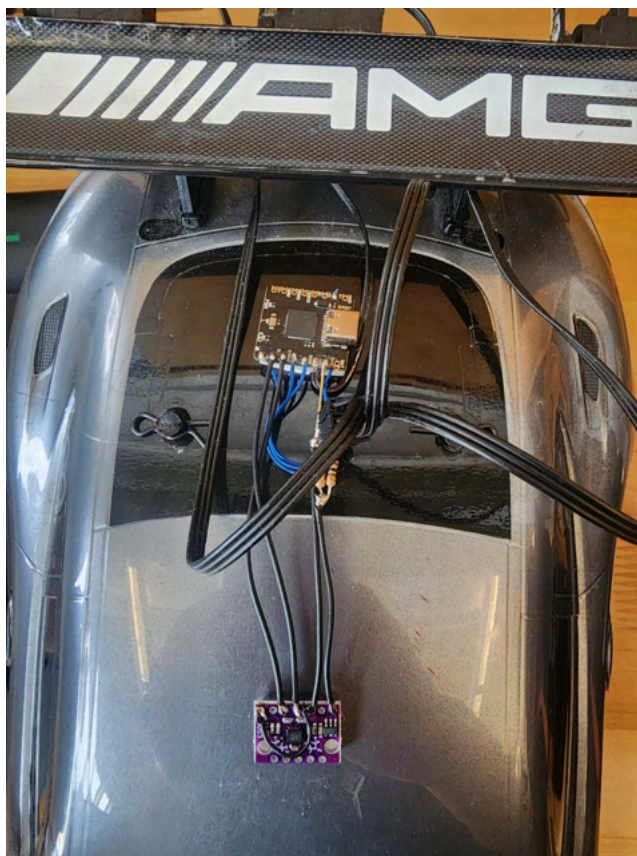
My work focused on identifying timing and signal integrity constraints, validating sensor behavior through testing and calibration, and refining the system to achieve stable and predictable operation. The resulting system demonstrated reliable engine position detection, consistent instrumentation readings, and embedded behavior suitable for integration into a larger vehicle control architecture.

Tools & Technologies

Embedded ECU, engine position sensors, custom reluctor wheel design, wiring harness design, signal conditioning, sensor calibration, telemetry & data logging, system testing & validation

Active Aerodynamic Control System (1/10 Scale Vehicle)

Closed-Loop Control & Sensor-Driven Actuation



K_p : Controller gain, for overall tuning of the system
 K_{s1} : gain for servo 1 = $Y_1(s) = G_{s1} \cdot U(s)$
 K_{s2} : gain for servo 2 = $Y_2(s) = G_{s2} \cdot U(s)$
 K_{s3} : gain for servo 3 = $Y_3(s) = G_{s3} \cdot U(s)$
 Overall output $\rightarrow Y(s) = Y_1(s) + Y_2(s) + Y_3(s)$

Feedback loop:
 $F(s) = G_{f1} \cdot Y_1(s)$
 $F(s) = G_{f2} \cdot Y_2(s)$
 $F(s) = G_{f3} \cdot Y_3(s)$
 $F(s) = G_{f1} \cdot Y_1(s) + G_{f2} \cdot Y_2(s) + G_{f3} \cdot Y_3(s)$
 $E(s) = U(s) - F(s)$
 $E(s) = U(s) - (G_{f1} \cdot Y_1(s) + G_{f2} \cdot Y_2(s) + G_{f3} \cdot Y_3(s))$
 $E(s) = U(s) - (G_{f1} \cdot G_{s1} \cdot U(s) + G_{f2} \cdot G_{s2} \cdot U(s) + G_{f3} \cdot G_{s3} \cdot U(s))$
 $E(s) = U(s) \cdot (1 - (G_{f1} \cdot G_{s1} + G_{f2} \cdot G_{s2} + G_{f3} \cdot G_{s3}))$
 Solving for $G(s)$
 $U(s) = G(s) \cdot E(s)$
 $G(s) = \frac{U(s)}{E(s)} = \frac{1}{1 - (G_{f1} \cdot G_{s1} + G_{f2} \cdot G_{s2} + G_{f3} \cdot G_{s3})}$
 Servo response: $Y(s) = G(s) \cdot U(s)$
 $\therefore G(s) = \frac{Y(s)}{U(s)} = \frac{G_{s1} \cdot G_{s2} \cdot G_{s3}}{1 - (G_{f1} \cdot G_{s1} + G_{f2} \cdot G_{s2} + G_{f3} \cdot G_{s3})}$

We know from PID tuning
 $K_p = 2$
 $K_i = 0.5$
 $K_d = 0.2$
 From opintuning
 $a_2 = K_p + T = 0.2 + 1$
 $a_1 = 1 + K_p = 1 + 2 = 3$
 $a_0 = K_i = 0.5$
 And we also know
 Time constant T_r in relation to dynamics of system in the servo speed in degree/s is 0.2
 We will use $G_c(s), G_f(s) = C(s) \cdot G_s(s)$
 For $C(s)$ poles
 $C(s) = K_p + K_i/s + K_d \cdot s$
 For term K_i has a pole @ $s=0$
 For $G_s(s)$ poles
 $\frac{1}{(0.2s+1)} = \frac{1}{0.2} = -5$ pole @ $s=-5$
 For $C(s)$ zeros
 $s = -\frac{K_p}{K_d} = -\frac{2}{0.2} = -10$
 For $G_s(s)$ zeros (numerator has no s coefficients)
 Poles @ $s=0, -5$
 Zeros @ $s=-10$
 No complex roots! -> no break in/out points
 Controller = $\frac{\sum \text{zeros}}{\sum \text{poles}}$
 $= \frac{(s+10) \cdot (-10)}{(s+5) \cdot s} = \frac{-10(s+10)}{s(s+5)}$
 Asymptote angle = $\frac{\sum \text{zeros} - \sum \text{poles}}{2-1}, K=0, 1, 2, \dots$
 $= 180^\circ$

Project Snapshot

Focus: Closed-loop control systems, sensor integration, real-time actuation

Scope: Actively controlled aerodynamic system developed for a 1/10 scale vehicle to study dynamic response during braking and cornering

My Ownership

I was responsible for the complete design, implementation, and validation of an active aerodynamic control system, spanning sensor selection, control logic development, and physical actuation.

- Design and implementation of closed-loop control logic for aerodynamic surfaces
- Integration of inertial sensors to measure vehicle dynamics
- Development of control strategies to coordinate multiple actuators based on vehicle state
- Servo actuation design and mechanical integration of aerodynamic elements
- System calibration, testing, and iterative refinement of control behavior

Key Technical Challenges

This project required translating abstract control concepts into a physically responsive system operating under real-world constraints.

- Interpreting inertial sensor data to derive meaningful control inputs
- Coordinating multiple actuators while maintaining stable and predictable response
- Managing latency and noise in sensor readings affecting control-loop behavior
- Debugging system behavior where failures could originate from sensors, control logic, or mechanical actuation

Engineering Summary

This project emphasized applying control theory to a real physical system where sensing, computation, and actuation are tightly coupled. Many challenges only emerged once the system was operating dynamically, requiring iterative tuning and validation to achieve stable behavior.

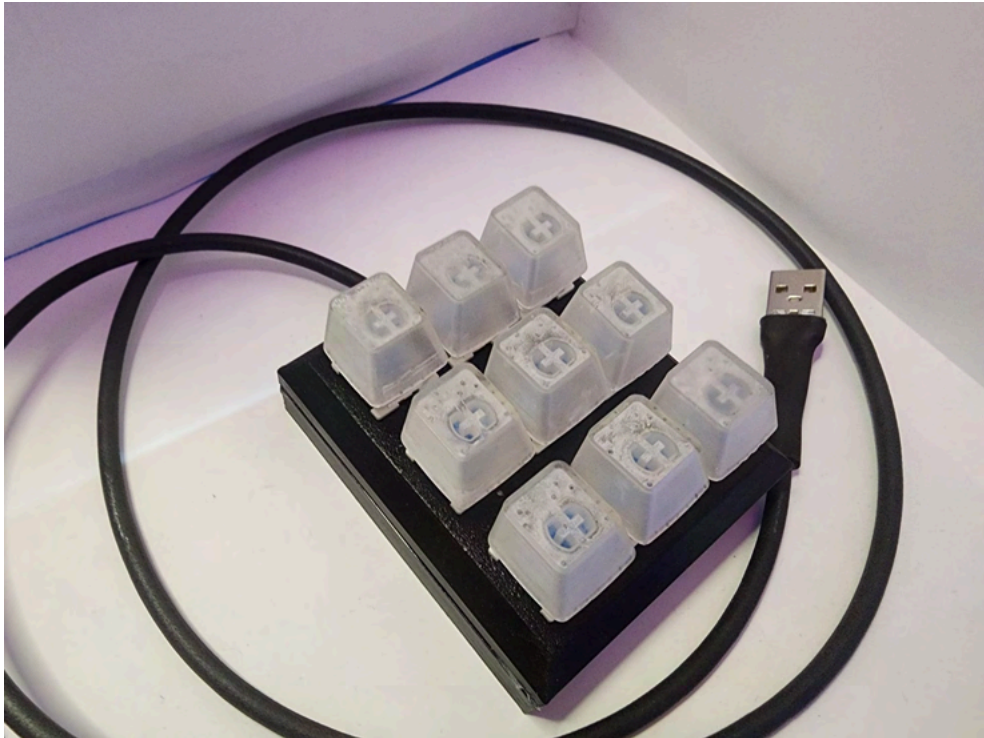
My work focused on ensuring that sensor inputs were reliable, control responses were predictable, and mechanical actuation accurately reflected control intent. The resulting system demonstrated responsive aerodynamic actuation driven by vehicle dynamics, providing hands-on experience in implementing closed-loop control systems beyond simulation.

Tools & Technologies

Embedded microcontroller, inertial sensors (IMUs), servo actuators, closed-loop control logic, mechanical integration, system testing & validation

Adaptive / Assistive Keyboard

Human-Centered Design & Accessible Hardware Development



Project Snapshot

Focus: Accessible hardware design, human-centered engineering, requirements-driven development

Scope: Custom adaptive keyboard developed to support users with physical or cognitive accessibility needs as part of a design-focused engineering course

My Ownership

I was responsible for the design and implementation of an adaptive keyboard solution, with an emphasis on usability, accessibility requirements, and practical hardware constraints.

- Translation of accessibility requirements into functional hardware design criteria
- Mechanical and electrical design of a custom input device
- Component selection and layout to support usability and reliability
- Assembly and validation of the physical prototype
- Documentation and presentation of design decisions to faculty evaluators

Key Technical Challenges

Designing an assistive input device required balancing technical feasibility with user-centered requirements, where usability considerations were as critical as electrical or mechanical performance.

- Interpreting accessibility-focused requirements into concrete engineering constraints
- Designing hardware that supported alternative interaction methods while remaining intuitive
- Balancing customization, robustness, and manufacturability
- Evaluating design effectiveness without direct end-user testing

Engineering Summary

This project emphasized a human-centered approach to engineering, where success was measured not only by technical correctness but by how effectively the design addressed user needs. The keyboard was developed through a structured design process that prioritized accessibility, clarity of interaction, and practical implementation constraints.

My work focused on translating abstract accessibility goals into a tangible hardware solution, validating functionality through prototype testing, and clearly communicating design tradeoffs. The project was evaluated by professors and teaching professionals as part of a design principles and project management course, reinforcing the importance of user-focused thinking in engineering practice.

Tools & Technologies

Custom hardware design, mechanical design, electronic components, prototyping & assembly, design documentation, system validation